

Beam Current Limits

USPAS - Beam Physics with Intense Space Charge

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1. Beam current limits in axisymmetric focusing channels
2. Beam current limits in quadrupolar focusing channels

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1.1 Radial envelope equations

For axially symmetric focusing systems, we derived the radial envelope equation:

$$\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta \rangle}{\gamma m\beta c}\right)^2\frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{\tilde{r}} = 0 \quad (1)$$

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + q r A_\theta, \quad (2)$$

$$\varepsilon_r \equiv \sqrt{\langle r^2 \rangle \langle r'^2 \rangle - \langle r r' \rangle^2 + \langle r^2 \rangle \langle r^2 \theta'^2 \rangle - \langle r^2 \theta' \rangle^2} \quad (3)$$

$$Q \equiv \frac{1}{mc^2 \beta^2 \gamma^3} \frac{q \langle \lambda(r) \rangle}{2\pi \epsilon_0} \quad (4)$$

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2.1 RMS envelope equations

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x(s)^2}{\tilde{x}^3} - \frac{2Q}{\tilde{x} + \tilde{y}} = 0, \quad (5)$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y(s)^2}{\tilde{y}^3} - \frac{2Q}{\tilde{x} + \tilde{y}} = 0. \quad (6)$$